

# Anshul Shah

[anshul7@uw.edu](mailto:anshul7@uw.edu) | [linkedin.com/in/anshul-shah1/](https://linkedin.com/in/anshul-shah1/) | [github.com/anshll](https://github.com/anshll) | [anshll.github.io/](https://anshll.github.io/)

## EDUCATION

### University of Washington

Seattle, WA

Bachelor of Science in Computer Science | GPA: 3.91

Sept 2023 – June 2027

**Coursework:** Distributed Systems, Machine Learning, Reinforcement Learning (grad), Operating Systems, Algorithms, Probability & Statistics, Data Structures & Parallelism, Linear Algebra, Differential Equations, Wireless Communication

## EXPERIENCE

### Cockroach Labs

June – Sept 2026

Software Engineer Intern

Key Value Team | New York, NY

- Extended CockroachDB's randomized serializability checker (KVNemesis) with fault injection to crash storage nodes in **our new disaggregated storage layer**, uncovering a **release blocking P0 bug** where writers never refresh replica liveness — causing **permanent loss of quorum** after transient failures despite a truly live majority — fixed **directly by the CTO**
- Further extended KVNemesis to generate new storage-specific Raft commands; with both enhancements, I found **10 new bugs**, including root-causing incorrect MVCC statistics to LSM compaction not preserving tombstone metadata.
- Extended CockroachDB's test framework so any test can run over a live disaggregated storage cluster instead of a mock; **scoped and handed off a design** for running the full test suite on a single shared cluster to amortize provisioning cost.

### Amazon Web Services

May – August 2025

Software Development Engineer Intern

Config Resource Onboarding Team | Seattle, WA

- Built a serverless, AI-powered system to automate onboarding new AWS resource types (RTs) to AWS Config — a **day-long manual process now 2 minutes**. Invoked via ApiGateway endpoint, outputs in Dynamo, S3
- Grounded Bedrock LLMs in RT documentation to author XML specs with per-resource-type dynamic tags/attributes and JSLT transformations for mapping “describe” API responses into standardized AWS Config configuration items.
- Created feedback loop with existing tests to let LLM self-correct until specs passed. On **30 RTs, 100% test accuracy**.

### Dartmouth Health

June – Sept 2024

Machine Learning Intern

Hanover, NH

- Assessed literature, designed, and trained a BERT based model for Named Entity Recognition (NER) of phenotypes in clinical notes with PyTorch — evaluated by a geneticist to automate **80% of manual annotation work**.
- Combined techniques from **state-of-art models PhenoBCBERT & PhenoID**: IOB labeling for more reliable annotations, window+striding for larger context, and 2nd finetuned BERT to normalize to phenotype IDs in ontology.
- Deployed Flask microservice/API to serve model, with a React **user-friendly annotation interface** for error correction

### Dubhacks NEXT

June 2024 – Present

Managing Director

Seattle, WA

- Lead UW's student-run startup incubator; our alumni have **raised \$5M cumulatively**, multiple at YCombinator
- Lead sponsorship outreach, secured **\$35K from Amazon, Hashicorp, PackVC, Madrona Ventures** over 2 years

## PROJECTS

### Distributed Key-Value Store | Java

Feb – March 2026

- Built a **linearizable KV store sharded** across Multi-Paxos replica groups with a dedicated controller for rebalancing
- Implemented **fault-tolerant** Paxos leader election, log slot proposals, merging, and garbage collection plus shard rebalancing logic; all messages **resilient** against simulated network partitions, drops, delays, and duplicates

### Tokens and Orders | Python, Torch, MuJoCo

April – June 2026

- Isolated the differences between autoregressive (AR) and diffusion generative control policies by training AR and masked diffusion (MD) policies within a common discretized action space and shared transformer backbone — to our knowledge the **first fair, controlled comparison**, and one of the **first explorations of MD models for robotic control**.
- Ran a factorial ablation showing **confidence-based decode order, not self-correction**, drives MD's advantage over AR
- Ablated multiple action tokenizers (naive binning, coarse-to-fine, bidirectional information distributions); MD outperformed AR across all of them, with results limited to a single benchmark task; [full paper here](#)

## TECHNICAL SKILLS

**Language/Framework:** Java, Python, C/C++, JavaScript/TypeScript, Go, HTML/CSS, Node.js, React, Svelte, Express  
**Technologies/Libraries:** Git, Docker, AWS, Unit Testing | NumPy, tensorflow, Torch, transformers, Matplotlib, scikit-learn